Understanding Retail Consumer Behavior: Predictors of Spending, Product Choice, and Customer Satisfaction

Alice Le, Michelle Jiang, Hailey Tan, Iris Yin

Motivation section (Iris)

What is the quality of these elements? Does the title tell us what to expect of the report to follow? Does the motivation section convince the reader that the broader study of this data set could elicit interesting insights?

Description of Dataset (Alice)

As stated, the dataset describes the shopping behavior of consumers, where each row represents an individual customer and their purchasing activity, including what they bought, how much they spent, how often they shop, their demographic attributes, and whether they used discounts or promotional codes.

Generally, the dataset is of good quality with no missing values or duplicate rows. There are 3900 rows in total, and each row corresponds to a unique customer's purchase transaction collected across a variety of U.S. states.

Where does it come from?? As stated, the dataset describes the shopping behavior of consumers, where [insert]

The variables in the dataset are:

- `Customer.ID`: The unique ID of each customer
- `Age`: The age of customer (in years). Range from 18 to 70 years old.
- `Gender`: Gender of customer
- `Item.Purchased`: Type of item that the customer bought, such as Blouse, Sweater, Scarf, etc.
- `Category`: Broader category of the item purchased. There are 4 categories: Clothing, Footwear, Outerwear, and Accessories
- `Purchase.Amount..USD`: The amount spent on the purchased item, in US dollars
- `Location`: The US state where the customer resides
- `Size`: Size of the clothing item. There are 4 sizes: S, M, L, XL
- `Color`: Color of the clothing item
- `Season`:
- `Review.Rating`
- `Subscription.Status`
- `Shipping.Type`: The method of shipping used for the order (e.g., "Express", "Free Shipping", "Next Day Air", "2-Day Shipping", "Store Pickup"). Indicates speed and delivery preferences
- `Discount.Applied`: Whether a discount was applied to the purchase
- `Promo.Code.Used`: Whether a promotion code was applied to the purchase
- `Previous.Purchases`: The number of purchases the customer made before this transaction
- `Payment.Method`: The payment method used for the transaction. There are 6 options: Cash, Bank Transfer, Debit Card, Credit Card, Venmo, and Paypal

- `Frequency.of.Purchases`: How often the customer shops. The frequencies include Weekly, Fortnightly, Bi-weekly, Monthly, Every 3 Months, Quarterly, and Annually

3 Research Questions

Stated below are our 3 research questions:

1. What factors predict how much a customer spends (Purchase Amount)?
2. How does season affect the types of items purchased and the color of the items?
3. How do discounts and shipping speed affect customer satisfaction (Review Rating)?

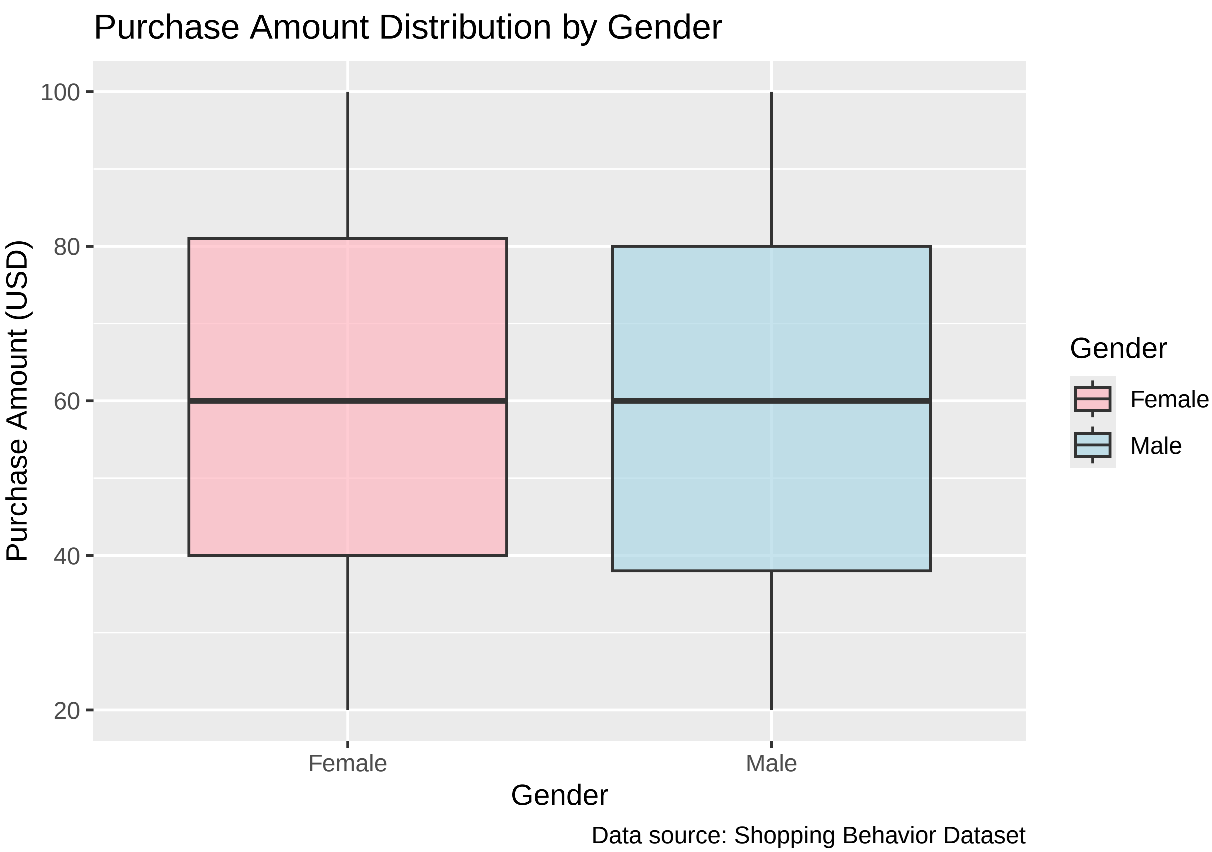
These research questions address core mechanisms of consumer behavior: spending determinants, seasonal purchasing patterns, and satisfaction drivers. Together, they allow us to examine how economic, temporal, and operational factors shape shopping outcomes in a way that is meaningful for both behavioral analysis and retail decision-making.

Question 1: What factors predict how much a customer spends (Purchase Amount)?

Boxplots - Hailey

To understand what drives customer spending, we start by examining how purchase amounts vary across key customer characteristics. Since purchase amount reflects a customer's level of financial engagement, exploring its relationship with demographic and behavioral factors can help reveal broader spending patterns that can inform future marketing, segmentation, and retail strategies.

As a first step, we look at gender as a core demographic variable. To compare spending across gender categories, we use a boxplot of purchase amount. This visualization highlights typical spending levels, the spread of purchase amounts, and differences in variability between groups, providing a clear and intuitive foundation for identifying factors that may predict customer spending.



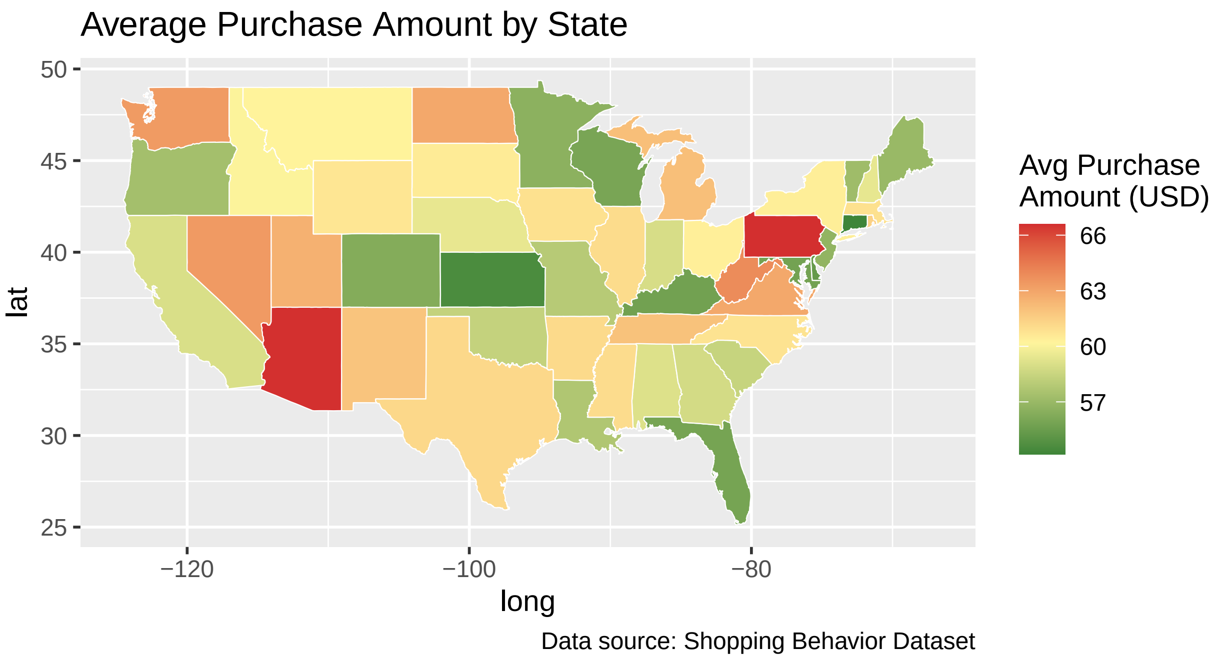
The boxplot shows that purchase amounts for males and females follow very similar patterns, with both groups centered around the mid-50 to 60 USD range and spanning from roughly the high 30s to the low 80s. The spreads and quartiles overlap substantially, indicating that gender might not be a strong predictor of how much customers spend. However, there is a small but noticeable difference in that the median purchase amount for females is slightly higher, suggesting they may spend marginally more on average, though not enough to indicate a major distinction between groups.

This comparison helps establish that gender might not meaningfully drive differences in spending behavior. And thus for our research question, this finding is relevant as it suggests we should look beyond basic demographics. As the distributions are nearly identical, gender-based segmentation would likely not improve predictions of customer spending.

Given the minimal differences observed here, our next step is to examine whether location provides stronger explanatory power as looking at purchase amount by geographic area may uncover regional spending patterns or market-level differences that gender alone cannot capture. The spatial graph below is especially effective since location-based trends are easier to interpret visually than in tables or non-spatial charts. By mapping average purchase amounts across the U.S., we can quickly identify clusters, regional contrasts, and potential spending hotspots.

Spatial Graph - Hailey

Average Purchase Amount by Location



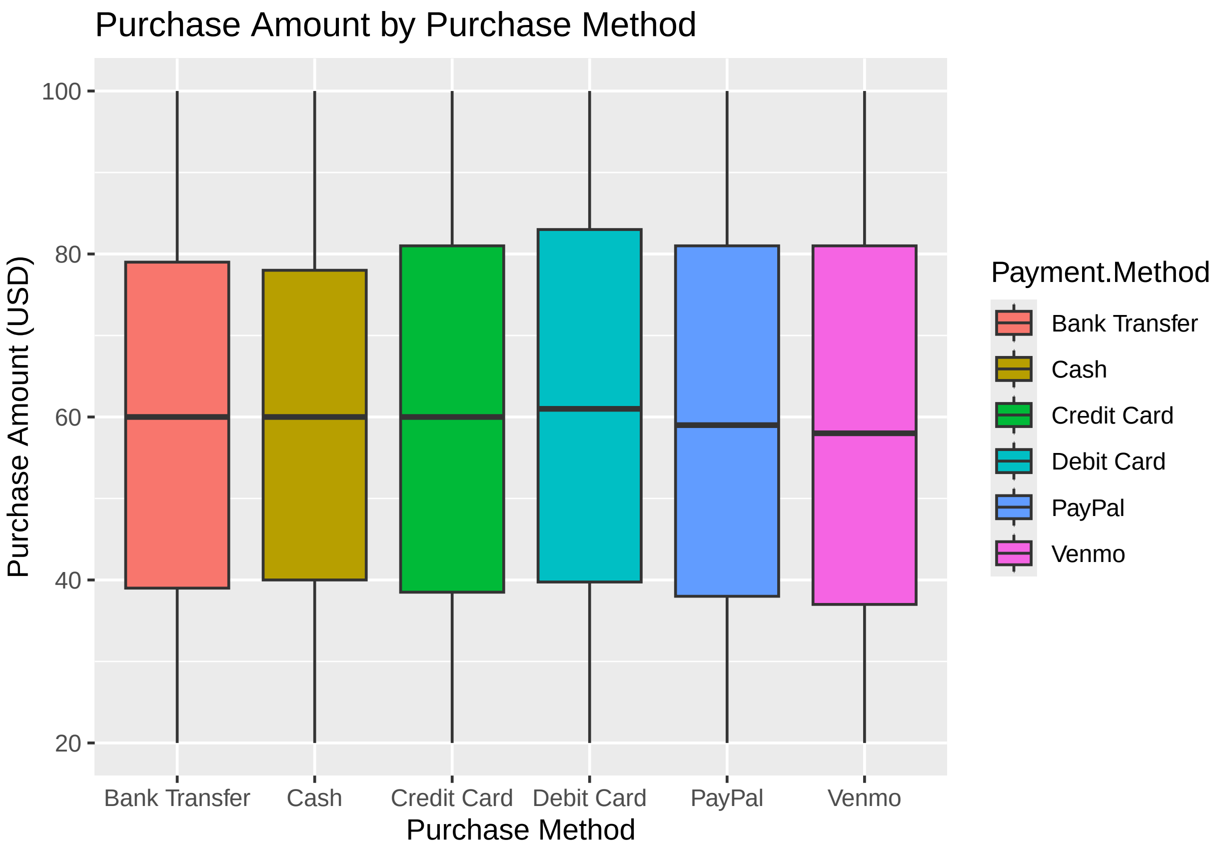
The spatial analysis graph reveals clear regional variations in customer spending across the United States, with darker green areas indicating lower averages (around 57 USD) and red areas showing higher spending (around 66 USD). New Mexico and Pennsylvania record the highest levels, followed by Arizona, Montana, and parts of the upper Midwest, while the central plains, southern states, and sections of the West Coast fall within a moderate 60–63 USD range. Higher spending in New Mexico likely reflects dispersed populations and affluent tourist influence, whereas Pennsylvania's pattern might stem from higher urban incomes and strong retail markets.

Understanding these trends can provide actionable insights for regional strategy. Retailers can focus premium products, loyalty programs, and targeted campaigns in high-spending markets while emphasizing value pricing and promotions in lower-spending areas. Beyond retail, such insights support more effective site selection, forecasting, and resource allocation.

Boxplot - Iris

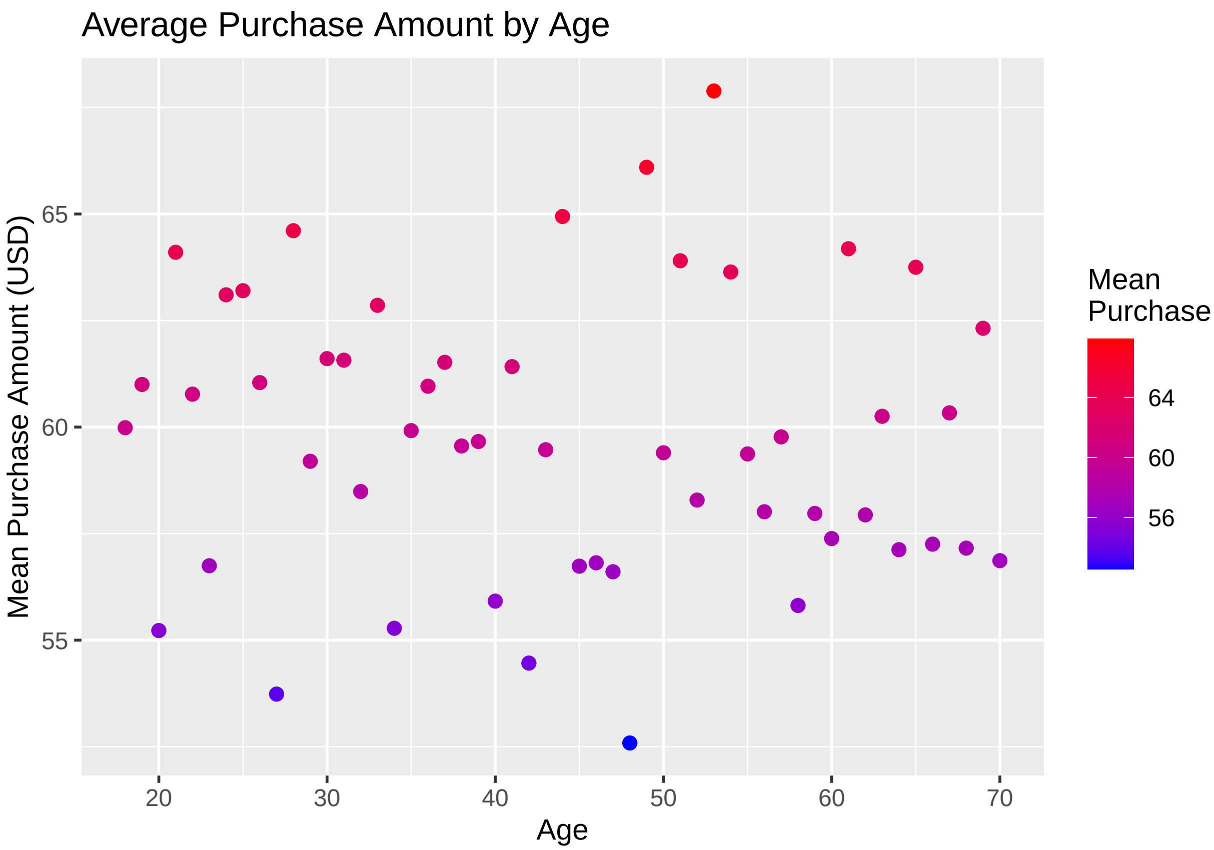
Correlations between:

Purchase Amount PUrchase method



Scatterplot - Iris

Purchase Amount vs Age



Average Purchase Amount by Age
Mean Purchase Amount (USD)
65
60
55
20
30
40
50
60
70
Age
Mean Purchase
64
60
56

Question 2: How does season affect the types of items purchased and the color of the items?

Season does not seem to have a significant affect on the types of items purchased and color of the items.

Variables involved: Season (Winter, Spring, Summer, Fall).Type of item purchased – e.g., Category or Item Purchased, Color of item – Color

Possible analyses:Use visualizations such as stacked/clustered bar charts or mosaic plots to compare item types and colors across seasons.

Chi-square test for association

Logistic regression (Predict "Promo Code Used = Yes/No")

Word Cloud - Michelle

color frequency by season

Heatmap - Michelle

This handles large numbers of categories beautifully.

Why it works:

Items go on the Y-axis

Seasons go on the X-axis

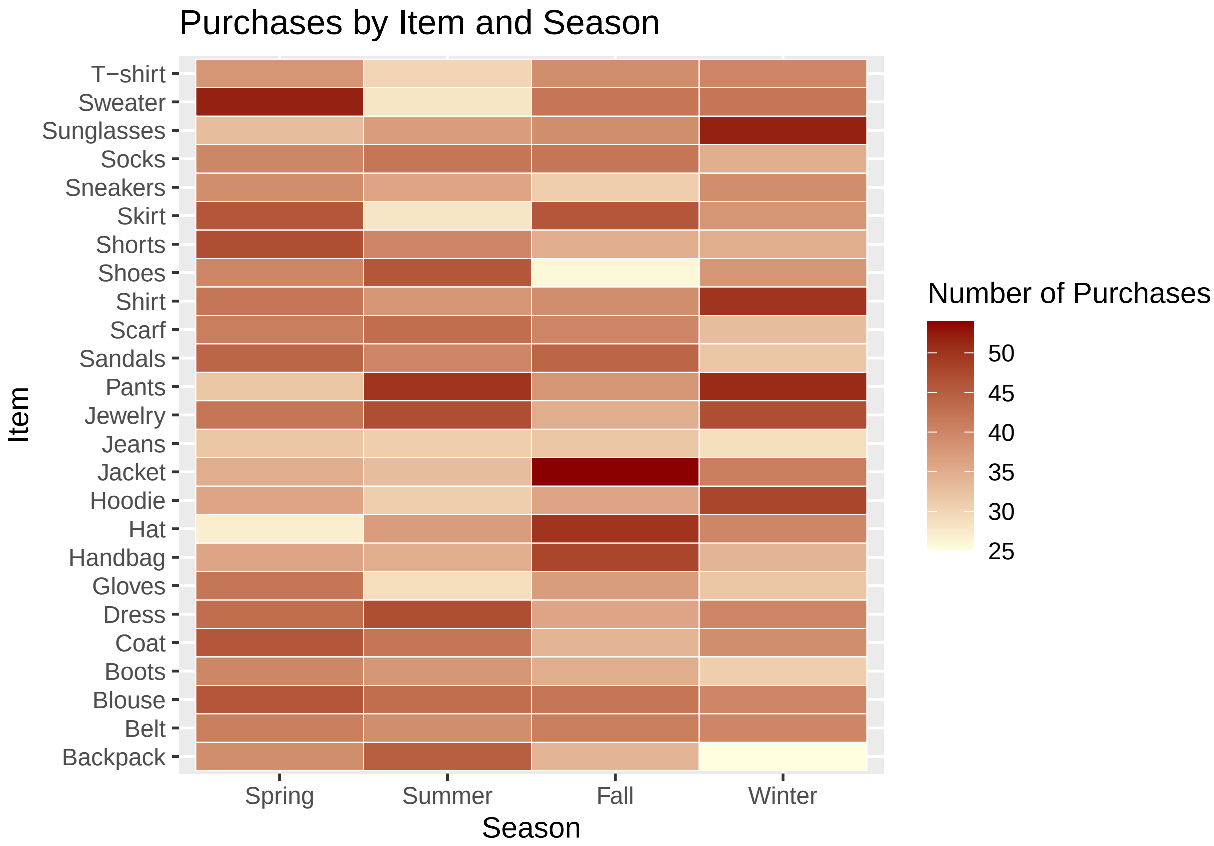
Color = number of purchases

It scales well even if you have 20–50+ item types.

Interpretation:

Darker = purchased more often

Lighter = purchased less often Frequency of Purchases (colored by subscription status) vs. purchase amount If frequency is truly numeric and continuous, this can show trends.



1. Chi-Square Test of Independence (Most appropriate)

Use this when both variables are categorical.

You can run: a. Season × Item Purchased

Tests whether certain items are bought more often in some seasons.

b. Season × Color

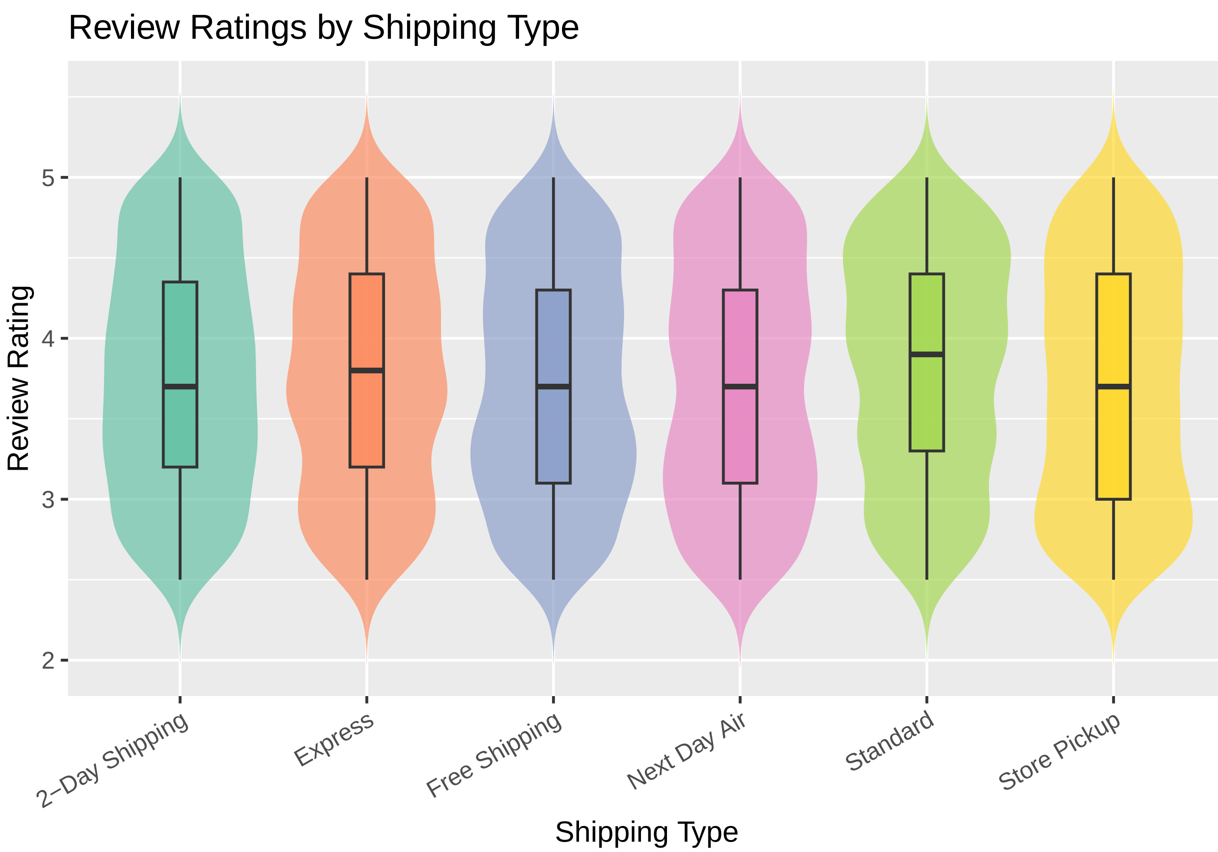
Tests whether certain colors are bought more often in certain seasons.

Question 3: How do shipping speed and discount affect customer satisfaction (Review Rating)?

Shipping speed and discounts are key components of the customer experience, and both may meaningfully influence how satisfied customers feel with their purchase. Hence, we would visualize the distribution of customer satisfaction across different shipping methods and whether there are discount applied using violin plots. Besides providing the the 5-number summary like what a box plot would, a violin plot also shows the distribution so we can examine whether there is skewness or outliers as well.

Then, we would conduct an ANOVA test to examine the mean review ratings across 6 different shipping methods and a two-sample t-test to compare review ratings between two groups ("Yes" vs. "No"). These tests would help us answer whether in this dataset, shipping speed and discount have a statistically significant relationship with customer satisfaction.

Review Raings by Shipping Type



From the following violin plot, we observe that the median rating are fairly similar across all shipping types, ranging between 3.6 to 3.7. Yet, standard and express shipping have a slightly higher median. The other values like min, max, first and third quartile seem to be similar as well. There's no outlier and skewness for any shipping type. All shipping types exhibit a similar spread, which suggests that variability in satisfaction may not strongly tied to delivery method.

```
##                 Df Sum Sq Mean Sq F value Pr(>F)
## Shipping.Type    5      6  1.2048   2.353 0.0384 *
## Residuals     3894   1994  0.5121
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Because the p-value for the ANOVA test is smaller than 0.05, we can conclude that average satisfaction differs across at least some shipping categories. However, when accompanying with the graph above, we see that there's not much variation in the distribution of average rating across shipping types. Hence, we can say that while shipping type may have a statistically detectable association with review ratings, the practical differences between shipping methods are likely minimal.

Review Raings vs. Discount Applied

The violin plot comparing review ratings for customers who did and did not receive a discount shows that the two groups have nearly identical distributions. Both groups center around ratings of approximately 3.7, with very similar spreads and shapes. There is no visible indication that receiving a discount leads to higher or lower satisfaction. The substantial overlap between the groups suggests little practical difference.

```
##
##  Welch Two Sample t-test
##
## data:  Review.Rating by Discount.Applied
## t = 0.77934, df = 3604.9, p-value = 0.4358
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -0.02737548  0.06349679
## sample estimates:
##  mean in group No mean in group Yes
##          3.757715          3.739654
```

The difference in mean ratings between customers who did not receive a discount (mean = 3.758) and those who did (mean = 3.740) is not statistically significant ($t = 0.779$, $p = 0.436$). The 95% confidence interval for the mean difference includes a zero (–0.027 to 0.063), indicating that any difference is small and likely due to random variation. Overall, the evidence suggests that discount usage does not meaningfully influence customer satisfaction.

From this dataset, we can conclude that the usage of discount does not affect rating. Even though the ANOVA test shows that there is statistical significance between shipping type and rating, the violin plots suggest there is not much variation.

Evaluation/Future work (Michelle)